

Averages and Spread

GCSE Maths · Worked Solutions

20 questions · 62 marks total

jtmaths.org · free maths resources

Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write down the mode of these numbers.

5 8 5 2 9

- 1 The mode is the most common value.
- 2 5 appears twice; every other value appears once.

Answer: 5

2 [2 marks]

Work out the mean of

3 9 4 8 6

- 1 $\text{Total} = 3 + 9 + 4 + 8 + 6 = 30$
- 2 There are 5 values, so the mean is $30 \div 5 = 6$

Answer: 6

3 [2 marks]

Here are six numbers.

14 7 11 3 9 20

- (a) Work out the median.
- (b) Work out the range.

- 1 Order them first: 3, 7, 9, 11, 14, 20
- 2 (a) With 6 values there is no single middle one, so average the 3rd and 4th.
- 3 $(9 + 11) \div 2 = 10$
- 4 (b) $\text{Range} = 20 - 3 = 17$

Answer: (a) 10 (b) 17

4 [2 marks]

A shoe shop wants to know which size to stock the most of.

Which average should the manager use, and why?

- 1 The manager wants the size that sells most often.
- 2 That is the mode — the most frequently occurring value.
- 3 A mean shoe size (such as 7.4) would not even be a real size to stock.

Answer: the mode — it gives the most commonly bought size

5

[3 marks]

The table shows the number of goals scored in 20 matches.

Goals	0	1	2	3
Frequency	5	8	4	3

Work out the mean number of goals per match.

1 Multiply each number of goals by its frequency.

2 $0 \times 5 = 0$, $1 \times 8 = 8$, $2 \times 4 = 8$, $3 \times 3 = 9$

3 Total goals = $0 + 8 + 8 + 9 = 25$

4 Total matches = $5 + 8 + 4 + 3 = 20$

5 Mean = $\frac{25}{20} = 1.25$

Answer: 1.25 goals per match

6

[3 marks]

The mean of four numbers is 12.

Three of the numbers are 8, 15 and 11.

Work out the fourth number.

1 If the mean of 4 numbers is 12, the total must be $4 \times 12 = 48$

2 The three known numbers add to $8 + 15 + 11 = 34$

3 Fourth number = $48 - 34 = 14$

Answer: 14

7

[3 marks]

Here is a set of data.

5 6 7 8 44

(a) Work out the mean.

(b) Work out the median.

(c) Which average better represents this data? Explain.

1 **(a)** Total = $5 + 6 + 7 + 8 + 44 = 70$, so the mean is $70 \div 5 = 14$

2 **(b)** The data is already in order, so the median is the 3rd value: 7

3 **(c)** The mean of 14 is larger than four of the five values, because the outlier 44 pulls it up.

4 The median of 7 is much more typical of the data, so it is the better average here.

Answer: (a) 14 (b) 7 (c) the median, because 44 is an outlier that distorts the mean

8

[3 marks]

Here are eight numbers in order.

3 5 7 8 11 13 16 20

(a) Work out the lower and upper quartiles.

(b) Work out the interquartile range.

1 The median splits the data into two halves of four values each: (3, 5, 7, 8) and (11, 13, 16, 20).

2 (a) The lower quartile is the median of the first half: $(5 + 7) \div 2 = 6$

3 The upper quartile is the median of the second half: $(13 + 16) \div 2 = 14.5$

4 (b) $\text{IQR} = 14.5 - 6 = 8.5$

Answer: (a) LQ 6, UQ 14.5 (b) 8.5

9

[4 marks]

The table shows the times, t minutes, taken by 40 people to complete a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	6	14	12	8

Work out an estimate for the mean time.

1 Use the midpoint of each class: 5, 15, 25 and 35.

2 Midpoint $\times$ frequency: $5 \times 6 = 30$, $15 \times 14 = 210$

3 $25 \times 12 = 300$, $35 \times 8 = 280$

4 Total = $30 + 210 + 300 + 280 = 820$

5 Mean = $\frac{820}{40} = 20.5$ minutes

Answer: 20.5 minutes

10

[4 marks]

Team A scored these goals in 8 matches: 0, 1, 1, 2, 3, 4, 4, 5

Team B scored these goals in 8 matches: 1, 2, 2, 2, 3, 3, 3, 4

(a) Work out the mean for each team.

(b) Work out the range for each team.

(c) Compare the two teams.

1 (a) Team A total = $0 + 1 + 1 + 2 + 3 + 4 + 4 + 5 = 20$, so the mean is $20 \div 8 = 2.5$

2 Team B total = $1 + 2 + 2 + 2 + 3 + 3 + 3 + 4 = 20$, so the mean is also 2.5

3 (b) Team A range = $5 - 0 = 5$ · Team B range = $4 - 1 = 3$

4 (c) The two teams score the same number of goals on average.

5 Team B has the smaller range, so its scoring is more consistent from match to match.

Answer: (a) both 2.5 (b) A: 5, B: 3 (c) same mean, but team B is more consistent

1 [2 marks]

The mean of 10 numbers is 18. Another number, 29, is added to the set.
Work out the new mean.

- 1 Total of the 10 numbers = $10 \times 18 = 180$
- 2 New total = $180 + 29 = 209$ over 11 numbers.
- 3 $209 \div 11 = 19$

Answer: 19

2 [2 marks]

The mean of five numbers is 14. Four of them are 10, 12, 15 and 18.
Work out the fifth number.

- 1 Total = $5 \times 14 = 70$
- 2 The four known numbers add to $10 + 12 + 15 + 18 = 55$
- 3 Fifth number = $70 - 55 = 15$

Answer: 15

3 [3 marks]

The mean mark of 8 boys is 65. The mean mark of 12 girls is 70.
Work out the mean mark of all 20 students.

- 1 You cannot average 65 and 70, because the groups are different sizes.
- 2 Boys total = $8 \times 65 = 520$
- 3 Girls total = $12 \times 70 = 840$
- 4 Overall total = 1360 over 20 students.
- 5 Mean = $\frac{1360}{20} = 68$

Answer: 68

4 [3 marks]

The table shows the times, t minutes, of 40 phone calls.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	7	15	13	5

Estimate the median time.

- 1 With 40 values the median is the 20th value.
- 2 Running totals: 7 up to 10 minutes, then $7 + 15 = 22$ up to 20 minutes.
- 3 So the 20th value lies in the class $10 < t \leq 20$, which holds 15 values.
- 4 It is the $20 - 7 = 13$ th value of those 15, so go $\frac{13}{15}$ of the way through a 10-minute class.
- 5 $\frac{13}{15} \times 10 = 8.67$, so the median is about $10 + 8.67 = 18.7$ minutes

Answer: about 18.7 minutes

5

[4 marks]

Here are nine values in order.

12 15 18 21 22 25 28 30 35

Work out the five values needed to draw a box plot, and the interquartile range.

- 1 Minimum = 12 and maximum = 35
- 2 With 9 values the median is the 5th: 22
- 3 The lower half (below the median) is 12, 15, 18, 21, so the lower quartile is $(15 + 18) \div 2 = 16.5$
- 4 The upper half is 25, 28, 30, 35, so the upper quartile is $(28 + 30) \div 2 = 29$
- 5 IQR = $29 - 16.5 = 12.5$

Answer: min 12, LQ 16.5, median 22, UQ 29, max 35; IQR = 12.5

6

[4 marks]

Box plots were drawn for two groups.

	Min	LQ	Median	UQ	Max
Group X	5	12	18	26	40
Group Y	8	14	17	21	30

- (a) Work out the interquartile range for each group.
 - (b) Make two comparisons between the groups.
- 1 **(a)** Group X: $26 - 12 = 14$ · Group Y: $21 - 14 = 7$
 - 2 **(b)** Averages: the medians are close (18 for X, 17 for Y), so typical values are similar, with X very slightly higher.
 - 3 Spread: Group Y has both a smaller interquartile range (7 against 14) and a smaller overall range (22 against 35), so its values are much more consistent.
 - 4 Always support a comparison with the numbers from the box plots.

Answer: (a) X: 14, Y: 7 (b) similar medians (18 v 17), but Y is far more consistent (IQR 7 v 14)

7

[4 marks]

The table shows the number of pets owned by some families.

Pets	0	1	2	3	4
Frequency	3	5	x	4	2

The mean number of pets is 1.8. Work out the value of x .

- 1 Total number of families = $3 + 5 + x + 4 + 2 = 14 + x$
- 2 Total number of pets = $0 + 5 + 2x + 12 + 8 = 25 + 2x$
- 3 Mean = $\frac{25 + 2x}{14 + x} = 1.8$
- 4 Multiply out: $25 + 2x = 1.8(14 + x) = 25.2 + 1.8x$
- 5 $0.2x = 0.2$, so $x = 1$
- 6 Check: 15 families, 27 pets, and $27 \div 15 = 1.8$ ✓

Answer: $x = 1$

8

[4 marks]

The table shows data grouped into classes of unequal width.

Value (x)	$0 < x \leq 5$	$5 < x \leq 15$	$15 < x \leq 35$
Frequency	8	22	20

(a) Work out an estimate for the mean.

(b) Work out which class is the modal class, explaining your reasoning.

1 (a) Midpoints are 2.5, 10 and 25.

2 $2.5 \times 8 = 20$, $10 \times 22 = 220$, $25 \times 20 = 500$, giving a total of 740

3 There are $8 + 22 + 20 = 50$ values, so the mean is $\frac{740}{50} = 14.8$

4 (b) With unequal widths you must compare frequency densities, not raw frequencies.

5 $8 \div 5 = 1.6$, $22 \div 10 = 2.2$, $20 \div 20 = 1.0$

6 The highest density is 2.2, so the modal class is $5 < x \leq 15$.

Answer: (a) 14.8 (b) $5 < x \leq 15$, because it has the greatest frequency density (2.2)

9

[4 marks]

A set of 12 numbers has a mean of 25.

Two of the numbers, 18 and 22, are removed.

Work out the mean of the remaining numbers.

1 Total of the 12 numbers = $12 \times 25 = 300$

2 The two removed numbers add to $18 + 22 = 40$

3 New total = $300 - 40 = 260$

4 There are now 10 numbers, so the mean is $\frac{260}{10} = 26$

5 The mean has gone up, which makes sense because both removed values were below 25.

Answer: 26

The table shows the number of siblings of 30 students.

Siblings	0	1	2	3	4
Frequency	6	10	8	4	2

- (a) Write down the mode.
(b) Work out the median.
(c) Work out the mean, to 2 decimal places.
(d) Write down the range.

1 (a) The largest frequency is 10, which belongs to 1 sibling, so the mode is 1.

2 (b) With 30 values the median is between the 15th and 16th values.

3 Running totals: 6 students have 0 siblings, then $6 + 10 = 16$ have 1 or fewer.

4 Both the 15th and 16th values are 1, so the median is 1.

5 (c) Total siblings = $0 \times 6 + 1 \times 10 + 2 \times 8 + 3 \times 4 + 4 \times 2 = 0 + 10 + 16 + 12 + 8 = 46$

6 Mean = $\frac{46}{30} = 1.5333\dots$, which is 1.53 to 2 decimal places.

7 (d) Range = $4 - 0 = 4$

Answer: (a) 1 (b) 1 (c) 1.53 (d) 4